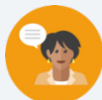


6.H2 Proofs by Contradiction

 Benchmark	MA.912.LT.4.8 Construct proofs, including proofs by contradiction.		 Learning Target	I can use proof by contradiction to justify that all circles are similar.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- What is a proof by contradiction? - Is there a relationship between the radius of circles and their circumference?			
 Lesson Narrative	This lesson introduces the idea of constructing a proof by contradiction by relating how an unlikely conclusion of a series of conditional statements can be proved as false using contradiction. Students then explore the relationships of similar circles to support student learning as they are guided through a proof by contradiction to prove that all circles are similar.			
 Component Summary	6.H2.1 Warm-Up (~5-10 min.)	Creating outrageous conditional statements (<i>SE p. 322</i>)	6.H2.3 Exploration (10 min.)	Discovering that corresponding parts of circles are proportional (<i>SE p. 323</i>)
	6.H2.2 Exploration (10 min.)	Writing contradiction statements (<i>SE p. 322</i>)	6.H2.4 Guided Instruction (15 min.)	Proving circles are similar using a proof by contradiction (<i>SE p. 324</i>)
	6.H2.5 Wrap-Up (~5 min.)	Reflection on understanding of the lesson (<i>SE p. 325</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Proof by Contradiction <u>Additional Terms:</u> - Corresponding - Proportional		- Commercial Video(s) (Optional) Index Cards for 6.H2.5 Wrap-Up	6.H2.1 Warm-Up contains a video to accompany the question prompt.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may write a contradiction as the inverse or converse. - Students may not understand that a proof by contradiction requires one to assume that a statement is true to then prove that the assumed statement is false.	- Students have learned that side lengths of corresponding similar figures are proportional but have not yet learned that other measures, like circumference of circles, share a proportional relationship. - Consider debriefing the 6.H2.2 Exploration before moving on to 6.H2.3 Exploration. This will allow the opportunity to ensure students were able to correctly write the negation statements in 6.H2.2 Exploration.
	Literacy and Language Routines	Mathematica; Thinking and Reasoning (MTR) Standard
	Notice and Wonder 6.H2.3 Exploration questions 1c and 1e already have the notice part of this routine. Consider implementing this routine by asking what the students may wonder. Making this move may get their curiosity piqued for other relationships of similar figures.	MA.K12.MTR.3.1: Mathematical Fluency This lesson is designed to lean on the students' mathematical fluency throughout the lesson. 6.H2.2 Exploration relies on students' ability to complete tasks accurately and with confidence while adapting procedures to a new context. 6.H2.3 Exploration relies on the students' abilities to maintain flexibility and accuracy while finding parts of a circle and ratios of parts of two different circles. Teachers are encouraged to provide opportunities for students to reflect on the method they used and determine if a more efficient method could have been used.
 Differentiate Instruction	The statements in the 6.H2.5 Wrap-Up provide an opportunity for students to monitor their own progress and guide their own effort and practice. Use responses to these prompts to work with students in developing a plan for moving toward mastery of these concepts.	☐ I understand how to write a contradiction. ☐ I need support on how to write a contradiction. ☐ I understand that all circles are similar. ☐ I need support on understanding why all circles are similar. ☐ I understand how to prove that all circles are similar by contradiction. ☐ I need support on how to prove that all circles are similar by contradiction.
Lesson Notes:		

6.H2.1 Warm-Up: Critical Thinking

Component Overview: Play the video of the commercial for the entire class. After students watch the video (31 seconds), prompt them to write five conditional statements.



6.H2.1 Warm-Up: Critical Thinking

1. DirecTV commercials are entertaining because the conditional statements are outrageous. Write five conditional statements with an outrageous conclusion that could be used for a DirecTV commercial.

Answers will vary.

1. *If your cable company has hidden fees, then you will end up swimming with sharks.*
2. *If your cable company charges for slowly increases their rates, then you will get locked out of your car.*
3. *If your cable company charges you more for less channels, then you will get charged by a rhinoceros.*
4. *If your cable company charges you a DVR service fee, then you will drink licorice flavored soda.*
5. *If your cable company has terrible customer service, then you will be chased by a heard of sheep.*



Allow students time to be creative with their five conditional statements. Establish the acceptable parameters so that students do not break any social boundaries in their statements. This is a great opportunity to engage students. However, it can easily take more than the designated amount of time. Consider assigning a student as a timekeeper to allow students to take ownership of the pacing of this lesson.

6.H2.2 Exploration: Contradictions

Component Overview: Students write contradiction statements to explore how contradictions change the outcome of conditional statements.



6.H2.2 Exploration: Contradictions

In the DirecTV commercial, there were seven conditional statements before the final conclusion.

Order	Statement
1	If your cable company puts you on hold, then you get angry.
2	If you get angry, then you go blow off steam.
3	If you go blow off steam, then an accident happens.
4	If an accident happens, then you get an eye patch from the doctor.
5	If you get an eye patch, then people think you are tough.
6	If people think you are tough, then they want to fight you.
7	If people want to fight you, then you get hurt.
8	You will end up in a roadside ditch.

1. Assume that the conclusion is not true. Write a contradiction to the conclusion.
You will not end up in a roadside ditch.
2. Share your contradiction with your partner.



6.H2.2 Exploration is scaffolded to allow students to engage with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the 6.H2.4 Guided Instruction. Students may revise their thinking throughout as they engage with each new question.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share with the whole group before 6.H2.4 Guided Instruction.

6.H2.2 Exploration and 6.H2.3 Exploration pair to set up students' understanding of how to use a proof by contradiction to prove that all circles are similar.

6.H2.2 Exploration: Contradictions (continued)

- Use your contradiction to write a "rewind" for the commercial with statements that would lead to the contradiction.

Order	Statement
8	<i>You will not end up in a roadside ditch.</i>
7	<i>If people will want to fight you, then you will not get hurt.</i>
6	<i>If people think you are tough, then they will not want to fight you.</i>
5	<i>If you get an eyepatch, then people will not think you are tough.</i>
4	<i>If an accident happens, then you do not get an eyepatch from the doctor.</i>
3	<i>If you go blow off steam, then an accident will not happen.</i>
2	<i>If you get angry, then you do not go to blow off steam.</i>
1	<i>If your cable company puts you on hold, then you do not get angry.</i>



Students may have a hard time following the reverse flow of these steps. The important thing here to focus on is the negation of each statement. Effectively writing negations is an important first step in writing proofs by contradiction later in the lesson.

- Explain how writing the contradiction would change the commercial.

Answers will vary. The commercial will not be very effective. If nothing bad happens when you get put on hold, like ending up in a ditch, then why should you care that you got put on hold?

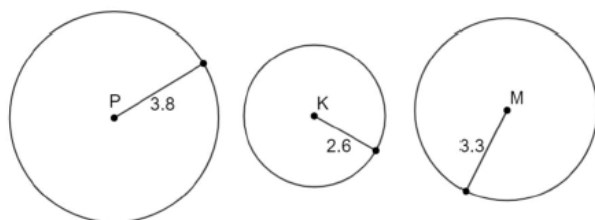
6.H2.3 Exploration: Circles Are Similar

Component Overview: Students will use the provided measurements to determine the circumference of the circles. They will then compare the ratio of the radius and circumference between the circles and write a conjecture based on their findings.



6.H2.3 Exploration: Circles are Similar

1. Circles P , K , and M are shown. The radius of Circle P is 3.8 units long. The radius of Circle K is 2.6 units long. The radius of Circle M is 3.3 units long.



- a. With your partner, fill in the table. Round to nearest thousandth. Use the π button on your calculator.

Circle	Radius	Circumference
P	3.8	23.876
K	2.6	16.336
M	3.3	20.735

- b. Complete the table.

Circle P and K	
Ratio of Radius of Circle P and Radius of Circle K	$\frac{3.8}{2.6} \approx 1.462$
Ratio of Circumference of Circle P and Circumference of Circle K	$\frac{23.876}{16.336} \approx 1.462$

- c. What do you notice about the ratios?
They are equivalent.



Consider assigning this as a group activity with groups of four.

- Assign groups roles.
- Complete question 1a as a group.
- Split groups into pairs.
- One pair completes questions 1b and 1c, the other completes questions 1d and 1e.
- Both pairs come together, share results, and work on question 2.

6.H2.3 Exploration: Circles Are Similar (continued)

d. Complete the table.

Circle <i>K</i> and <i>M</i>	
Ratio of Radius of Circle <i>K</i> and Radius of Circle <i>M</i>	$\frac{2.6}{3.3} \approx 0.788$
Ratio of Circumference of Circle <i>K</i> and Circumference of Circle <i>M</i>	$\frac{16.336}{20.735} \approx 0.788$

e. What do you notice about the ratios?

They are equivalent.

2. Make a conjecture about the ratios of radii and circumferences of circles.

Answers will vary. I think the ratios of the radii and the ratios of the circumferences of two circles will be equivalent to each other.



Notice and Wonder

Questions 1c and 1e already have the notice part of this routine. Before moving to question 2, consider asking what the students may wonder. Take a moment to explore any named wonderings about other ratios of corresponding parts of circles, like their areas.

6.H2.4 Guided Instruction: Proof by Contradiction

Component Overview: Students are introduced to a new form of proof called proof by contradiction. Using proof by contradiction, students determine that circles are similar.



6.H2.4 Guided Instruction: Proof by Contradiction

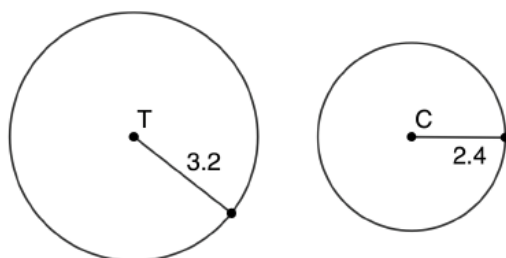
A way to write a proof is by using a contradiction.



Proof by Contradiction

Assume the statement to be proven is false, and work to show its falsity until the result of that assumption is a contradiction.

1. Circles T and C are shown. The radius of Circle T is 3.2 units long. The radius of Circle C is 2.4 units long.



Prove that Circle T is similar to Circle C .

- a. Write an assumption that the statement is false.
Assume that Circle T is not similar to Circle C .
- b. Find the ratio of the radius of Circle T to the radius of Circle C .
 $\frac{3.2}{2.4} \approx 1.333$
- c. Find the ratio of the circumference of Circle T to the circumference of Circle C .
 $\frac{6.4\pi}{4.8\pi} \approx 1.333$



Students naturally understand that all circles are similar so proving that two circles are similar may appear to be pointless. This is why proving that two circles are similar is a nice avenue for proof by contradiction.



Consider extending this component to include radii represented as variables and writing the circumference in terms of π . Prompt students to make connections between the ratios formed by the radii and the ratios formed by the circumference in this form. Then have students write the ratios of the areas of the circles and make a conjecture.

6.H2.4 Guided Instruction: Proof by Contradiction (continued)

- d. Explain how the ratios show the assumption is not true.

The radius and the circumference are proportional, so the circles are similar.

- e. Complete the proof.

Assume that Circle T is not similar to Circle C . The ratio of the radius of Circle T to the radius of Circle C is $\frac{4}{3}$. The ratio of the circumference of Circle T to the circumference of Circle C is $\frac{4}{3}$. The ratio of the radii and the ratio of the circumferences are equivalent, so corresponding parts of Circle T and Circle C are proportional. Since the corresponding parts of Circle T and Circle C are proportional, the circles are similar. This is a contradiction, so the assumption that Circle T is not similar to Circle C is not true.



Provide an opportunity for students to make connections between the contradiction and the original proof statement by asking, “Because the assumption that Circle T is not similar to Circle C is not true, what have we proved?” (Sample answer: that Circle T is similar to Circle C .)

6.H2.5 Wrap-Up: Reflection

Component Overview: Students complete a self-reflection on their level of understanding of the content of this lesson.



6.H2.5 Wrap-Up: Reflection

1. Check below all the statements that apply to your understanding.

Answers will vary.

- ☐ I understand how to write a contradiction.
- ☐ I need support on how to write a contradiction.
- ☐ I understand that all circles are similar.
- ☐ I need support on understanding why all circles are similar.
- ☐ I understand how to prove that all circles are similar by contradiction.
- ☐ I need support on how to prove that all circles are similar by contradiction.



Consider asking students to write their answers on a note card and drop them in a bucket on the way out of the class. Tally how many checks are given to each statement to determine where the most support is needed. Consider statements that have a large number of checks as a possible station on another day.